

| <b>(c) Genetic modification (genetic engineering)</b> |                                                                                                                                            |
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| <b>Students should:</b>                               |                                                                                                                                            |
| 5.12                                                  | understand how restriction enzymes are used to cut DNA at specific sites and ligase enzymes are used to join pieces of DNA together        |
| 5.13                                                  | understand how plasmids and viruses can act as vectors, which take up pieces of DNA, and then insert this recombinant DNA into other cells |
| 5.14                                                  | understand how large amounts of human insulin can be manufactured from genetically modified bacteria that are grown in a fermenter         |
| 5.15                                                  | understand how genetically modified plants can be used to improve food production                                                          |
| 5.16                                                  | understand that the term transgenic means the transfer of genetic material from one species to a different species                         |